Course: Data Collection & Preparation
Deliverables: Jupyter Notebook (.ipynb) + Writing report

Deadline for submitting: 10.10.2026
Deadline for defense: 14.10.2026
Maximum point - 7

Before starting the project, please send your instructor:
• your selected project topic;
• the data sources, tools, or technologies you plan to use;
• the names of your group members.

You may work individually or in a group of up to 4 students.

Deadline for sending this information: 16.09.2026.

Project Goal

The purpose of this project is to apply the complete data collection and preparation pipeline learned in class:

- Collecting real data from **two different sources** – one **API** and one **static web page** (via web scraping)
- Cleaning, merging, and transforming these datasets
- Performing basic **exploratory data analysis (EDA)**
- Creating **visualizations** to communicate insights

You will demonstrate your understanding of **data collection, preparation, and analysis** of real-world data.

You are free to choose your own topic. The example topic "Weather & Tourism Trends" (using Open-Meteo API and Wikipedia pages) is provided as an illustration of what a complete project might look like, but you may select any other domain of interest, such as education, economy, sports, culture, or environment.

Example Topic: Weather & Tourism Trends

Explore how climate (average temperature, precipitation) relates to tourism activity (number of tourist attractions listed on Wikipedia).

1. Retrieve weather data using the Open-Meteo API.
2. Collect tourism data by scraping Wikipedia category pages such as 'https://en.wikipedia.org/wiki/Category:Tourist_attractions_in_<City>'.
3. Clean, merge, and analyze your datasets in pandas.

4. Produce at least 3 visualizations with Matplotlib or Seaborn.
5. Summarize your findings in a short-written report.

Technical Requirements

Component	Description
API	Use an open API (e.g., Open-Meteo, World Bank, SpaceX, etc.) to collect structured data.
Web Scraping	Use requests + BeautifulSoup to extract HTML data (respect robots.txt).
Data Cleaning	Handle missing values, duplicates, datatypes, and renaming columns.
Merging	Combine data sources logically (e.g., by city, country, or date).
Analysis	Compute descriptive statistics and correlations.
Visualization	At least 3 clear and labeled plots (Matplotlib or Seaborn).
Report	1–2 pages describing approach, results, and key insights.

Submission Rules

Any commits or changes after this time will result in **disqualification** – the project will **not be accepted,** and you will receive **zero points**.

Make sure your repository is **public** or that your instructor has access.

The repository must contain:

- SIS_<TeamName>_Weather_Tourism.ipynb (example)
- report_<TeamName>.pdf
- requirements.txt (optional, if you used external libraries)

Oral Defense

Each pair will present their project during the defense:

- Duration: **~10 minutes per team**
- All team members must participate. If one of the members is not present, he/she will receive **zero points**.
- Be ready to answer questions about your code, API usage, scraping, and data logic.

Academic Integrity Policy

Each team must work independently and produce their own code and analysis.

The following actions are **strictly prohibited**:

- Copying or sharing code with other teams
- Using **AI tools** (e.g., ChatGPT, Copilot, Gemini, etc.) to generate code, text, or visualizations
- Using third-party scripts, notebooks, or templates not created by your team

All submissions will be reviewed for **similarity and originality**.

Students may be asked to explain their code during the oral defense.

If you cannot explain your own code, your project will receive **zero points**. If you didn't write it yourself, you must be ready to lose the points yourself.

Failure to read or correctly interpret this assignment description **does not exempt students from responsibility**.
It is each team's duty to carefully review all project requirements, deadlines, and rules. Excuses such as "*I didn't see that part"* or *"I misunderstood the instructions"* will **not be accepted**.